National University of Computer and Emerging Sciences,
Lahore Campus

	Course Name:	**Computer Networks**	**Course Code:**	**CS 3001**
	Program:	**BS (Software Engineering)**	**Semester:**	**Spring 2026**
	Duration:	**15 minutes**	**Total Marks:**	**15**
	Paper Date:	**3-February-2026**	**Section**	**6A , 6B**
	Exam Type:	**Quiz 1 - Chapter 1**	**Page(s):**	**2**

Student Name **Roll No.** **Section:**

Q1. <u>Encircle the correct option:</u> **[5 marks] [CLO 1]**

1. What does it mean for a network to achieve a "high utilization"?
   a. The network maintains low latency.
   b. The network fully uses available bandwidth.
   c. Packets are evenly distributed across all routes.
   d. The network ensures no packet loss occurs.

2. Nodal processing delay involves all of the following except:
   a. Checking for bit errors.
   b. Determining the output link.
   c. Waiting for the link to become idle.
   d. Examining the packet header.

3. The Bandwidth-Delay Product (BDP) effectively measures:
   a. Maximum number of bits present on the wire at any instant.
   b. The total time it takes for a packet to traverse the link.
   c. The speed at which bits are pushed onto the medium.
   d. The queuing delay experienced by the last bit.

4. Consider a path with three links with rates R1 = 200 kbps, R2 = 5 Mbps, and R3 = 1 Mbps. If a server transmits a large file to a client over this path, what is the theoretical maximum throughput?
   a. 5 Mbps
   b. 2.06 Mbps (Average)
   c. 1 Mbps
   d. 200 kbps

5. A ________ is formed at the transport layer by encapsulating an application layer message with transport layer headers.
   a. Segment
   b. Datagram
   c. Message
   d. Frame

Q2: Two hosts are connected via a packet switch with 10^6 bits per second links. The packet switch implements store-and-forward scheme, and each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds (time incurred on processing) after it receives that packet. If 10,000 bits of data are to be transmitted between the two hosts using a packet size of 5,000 bits, ignoring any queuing delays, please determine the time required to receive 10,000 bits at the destination. **[10 Marks] [CLO 1]**